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Economics of biofuels and bioproducts from an integrated pyrolysis biorefinery

Qi Dang, Wenhao Hu, Marjorie Rover, Robert C. Brown, Mark M. Wright, Iowa State University, Ames, Iowa, USA

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Abstract: This paper describes a novel integrated biorefinery based on fast pyrolysis and subsequent bio-oil upgrading to produce three product portfolios: hydrocarbon biofuels (gasoline and diesel), bio-based products (bioasphalt, calcium acetate, and dextrose), and hydrocarbon chemicals (aromatics, olefins, and fuel oil). Based on the economic assumptions and historical market prices of the products, we calculate internal rates of return (IRRs) of the biorefinery (2000 dry metric tons per day) over 30 years to evaluate its performance. Given the historical low, average, and high prices of the products, a total of 27 scenarios are explored and the maximum IRR for each scenario is determined. Results indicate that upgrading bio-oil to bio-based products yields the maximum IRR of 67.9% if bio-based products can be marketed at a historical high price level (between $257 and $1491 per metric ton) and an IRR of 42.5% at an average price level (between $240 and $950 per metric ton). Under some market conditions, the biorefinery should produce hydrocarbon chemicals or biofuels to increase the IRR. We also estimate the IRR probability distribution using Monte Carlo simulation by incorporating distributions of product prices and bio-oil fractions allocated for three product portfolios. Average IRRs of –3.9% for biofuels, 42.7% for bio-based products, and 17.3% for hydrocarbon chemicals are observed with standard deviations of 24.4%, 24.1%, and 6.5%, respectively. The integrated biorefinery exhibits an IRR distribution of 16.2%±16.4% suggesting that the integrated biorefinery could balance the economic disadvantages and benefits from individual product portfolio to achieve profitable IRRs. © 2016 Society of Chemical Industry and John Wiley & Sons, Ltd

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Key words: pyrolysis biorefinery; biofuels; bio-based products; hydrocarbon chemicals; techno-economic analysis

Corresponding author: Mark M. Wright, 2078 Black Engineering Building, Iowa State University, Ames, Iowa 50011, USA.
E-mail: markmw@iastate.edu

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Introduction

Biomass resources are abundant and widely distributed across the USA. As a promising renewable alternative, biomass plays an important role in enhancing the nation's energy security and mitigating carbon emissions. The *Billion Ton Study* predicts that approximately 30% of US petroleum consumption could be replaced with biomass resources by sustainably recovering one billion dry tons of biomass annually, primarily from agriculture and forest residues.[1] For the sake of reducing carbon emissions, replacing fossil fuels with carbon-neutral biomass resources is vital to achieving environmental targets.

Biomass can be thermochemically converted to a variety of fuels and products through gasification, combustion, and pyrolysis.[2–4] Fast pyrolysis, a rapid thermal decomposition of organic compounds, is an emerging advanced biomass conversion technology. It usually takes place at a moderate temperature of around 500°C in the absence of oxygen with a short hot vapor residence time of a few seconds.[5] The process yields a mix of liquid, gas, and solid products. The product distribution and its composition vary mostly with feedstock properties, pyrolysis temperature, and residence time. The primary liquid product, known as bio-oil, consists of hundreds of oxygenated compounds, which leads to several undesirable properties such as low heating value, high acidity, high viscosity, and immiscibility with petroleum fuels. Therefore, the direct application of bio-oil has been limited to the turbine, engine, and boiler units for heat and power generation.

Bio-oil is a platform intermediate product which can be upgraded into a variety of higher value products such as drop-in transportation fuels, infrastructure materials, and commodity chemicals. Elliott *et al.*[6] upgraded bio-oil into alkenes and aromatics through catalytic hydroprocessing followed by hydrocracking, which is a common application for drop-in transportation biofuel production. They also investigated hydroprocessing of phenolic oils derived from pyrolysis for hydrocarbon product production.[7] Peralta *et al.*[8] developed a method to produce bio-binder by combing bio-oil and rubber. It can be used as a modifier when added to petroleum-derived asphalt for pavement material application. Zhang *et al.*[9] evaluated the production of monosaccharide along with hydrogen and transportation fuels via biomass fast pyrolysis. Vispute *et al.*[10] found that bio-oil can be upgraded into industrial commodity chemicals through catalytic hydroprocessing and subsequent fluid catalytic cracking (FCC).

Although advanced biofuel production technologies using bio-oil have not been commercialized at large scale, biomass-derived products could replace their fossil-fuel-based counterparts if they can be produced in a cost-effective and sustainable way. Techno-economic analysis (TEA) provides a useful way to explore the economic feasibility of biomass conversion pathways towards commercialization.[11] Several studies have investigated the economics of various pathways from fast pyrolysis platform. Wright *et al.*[12] examined a fast pyrolysis facility that converted corn stover to naphtha- and diesel- range fuels. The product price was estimated to be $3.09 and $2.11 per gallon of gasoline equivalent (GGE) for internal hydrogen production and merchant hydrogen purchase cases, respectively. Li *et al.*[13] investigated the cost uncertainty of *in situ* and *ex situ* catalytic pyrolysis for transportation biofuel production. Minimum fuel selling prices (MFSPs) of $1.11 and $1.13 per liter were projected for these two emerging technologies. Brown *et al.*[14] evaluated the economic feasibilities of five different pathways that produced commodity chemicals using integrated catalytic processing (ICP) based on Vispute's work.[10] Hu *et al.*[15] compared the economics of three individual product portfolios from a fast pyrolysis biorefinery: biofuels, biochemicals, and hydrocarbon chemicals. They reported minimum product selling prices (MPSPs) of $0.82 per liter for biofuels, $433.7 per metric ton for biochemicals, and $773.5 per metric ton for hydrocarbon chemicals. Previous techno-economic analysis studies have evaluated and compared fast pyrolysis facilities focused on a single primary product, for example, either transportation fuels or commodity chemicals. Hydrogen was usually generated with drop-in transportation fuel production in order to produce internal hydrogen for bio-oil hydroprocessing. Most of the previous studies assume fixed product prices; however, these are associated with high uncertainties resulting from market conditions.

One potential approach to reducing the risks of biorefineries is to enhance their capabilities to produce diverse products depending on various market situations. Product selection and optimization of biorefineries with multiple products have been investigated by some researchers.[16–18] In this work, we introduce a novel conceptual biorefinery integrating biomass fast pyrolysis and three different bio-oil upgrading pathways. Bio-oil from fast pyrolysis is recovered into five different stage fractions through a unique bio-oil fractionation system developed by Iowa State University.[19] Three bio-oil fractions are simultaneously upgraded to drop-in transportation biofuels (gasoline and diesel), bio-based products (bioasphalt, calcium acetate, and dextrose), and hydrocarbon chemicals (aro-

matics, olefins, and fuel oil). The objective of this paper is to investigate the economics of the integrated biorefinery with diverse target product choices. We integrate data from various sources such as an Aspen Plus^{TM} process model, literature, and historical data into the economic model. We develop a new method to explore the internal rates of return (IRRs) for various combinations of three product portfolios and identify the maximum IRRs under given market conditions of product prices. Furthermore, we address the IRR uncertainty using Monte Carlo simulation by incorporating the uncertainties of product prices and bio-oil fractions allocated for upgrading in three product portfolios. The findings from this analysis will shed some light on product selection and risk mitigation for investors from the bioproduct industry.

Methodology

Process description

The integrated biorefinery is based on a conceptual level process design and assumed to have a biomass processing capacity of 2000 metric tons per day (MTPD). Red oak is the feedstock and its properties can be found in the previous paper.[15] The overall process consists of pre-treatment, pyrolysis, solid removal, biochar combustion, bio-oil fractionation, and bio-oil upgrading units as illustrated in Fig. 1.

The pre-treatment step starts with chopping red oak to a particle size of approximately 10 mm diameter. Acid treatment is adopted to increase the sugar yield as suggested by experiments. In this case, 0.4 wt% of sulfuric acid is infused onto biomass in a continuous spray drum after chopping.[9] Biomass is then dried in a steam rotary dryer to reduce the moisture content from as-received 25 wt% to less than 7 wt%. Finally, the red oak particle is further grounded to a diameter of around 3 mm.

Fast pyrolysis occurs in a fluidized bed pyrolyzer at 500°C. The hot vapor stream first goes through a clean-up unit where a high efficiency cyclone (95%) is used to remove particulate matter. Biochar, ash, and alkali metals are carried along in the solid stream. In this study, a portion of the biochar (around 52%) is combusted to provide steam and heat for biomass pre-treatment and pyrolysis while the excess biochar is sold as a byproduct from the biorefinery to increase its revenue. The clean vapor phase then flows through a novel five-stage bio-oil fractionation system. The fractionation system consists of a series of heat exchangers and electrostatic precipitators that condense bio-oil as five stage fractions (SFs) with distinctive

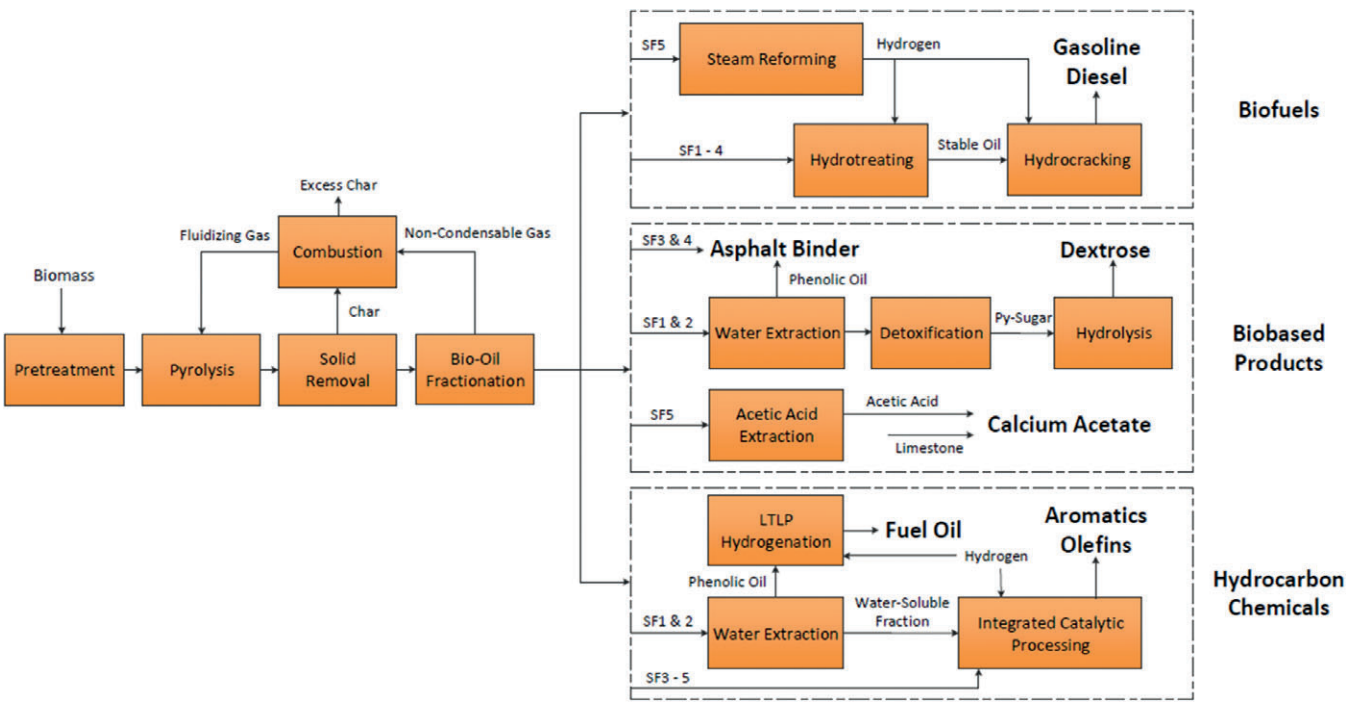

Figure 1. Schematic diagram of bio-oil production and upgrading for biofuels, bio-based products, and hydrocarbon chemicals production.

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properties. SF1 and SF2 are designated as heavy end fractions and mainly include phenolic oil and sugars such as levoglucosan. Monomeric phenols and furans are captured in the middle end fractions, SF3 and SF4. Water and light oxygenated compounds such as acetic acid are collected as light ends in SF5. Non-condensable gas (NCG) is recycled and combusted along with some biochar to provide heat for pyrolysis reactor system. The product yields from fast pyrolysis are based on experiments by Rover *et al.*[20]

The distinctive compositions of recovered stage fractions make them suitable for upgrading to various products. As shown in Fig. 1, three product portfolios are investigated in this study: biofuels, bio-based products and hydrocarbon chemicals. The whole bio-oil can be either upgraded to produce one product category or upgraded to a mixture of the three product categories. By varying bio-oil fractions targeting each product portfolio, a wide range of cases can be explored to determine the most economical way of operating the integrated biorefinery.

In the biofuels portfolio, the combination of heavy end and middle end fractions (SF1 to SF4) are stabilized through a two-stage hydrotreating step.[21] The first stage reaction occurs at 140°C with a pressure of 121 bar while the subsequent reaction takes place under a more severe condition of 370°C and 121 bar. Hydrogen required for hydrotreating is sufficiently provided by steam reforming of light ends SF5 with additional natural gas input.[12, 21] The stabilized bio-oil is then distilled to recover gasoline and diesel blendstock. The heavy bottoms from the distillation tower are further hydrocracked into gasoline and diesel blendstock. The overall conversion pathway of the biofuels portfolio including the upstream pre-treatment and fast pyrolysis is simulated using Aspen Plus™. Operational conditions are selected based on research data from experiments. The yields of gasoline and diesel are calculated from the process model.

In the bio-based products portfolio, asphalt binder, dextrose, and calcium acetate are produced as target products. The experiments suggest that most of the water-soluble sugars from fast pyrolysis stay in the heavy end fractions SF1 and SF2. Water extraction is used to separate sugars from phenolic oil in the heavy ends.[22] A subsequent detoxification step commonly by over-liming or resin separates pyrolytic sugars with high concentrations. The clean pyrolytic sugar consists mostly of monosaccharides, levoglucosan, and some disaccharides.[22] The sugars are hydrolyzed in 400 mM sulfuric acid to produce a glucose-rich syrup, which is designated as dextrose in our analysis. The phenolic oil from the water extraction, consisting of phenolic monomers and oligomers with properties similar to petroleum-based bitumen, is used to produce bioasphalt binder. Peralta *et al.*[8] have reported the use of similar pyrolysis-derived compounds in bio-binders for asphalt applications. We assume the bioasphalt binder is sold at the same price as petroleum-derived asphalt. The middle end fraction, consisting primarily of monomeric phenols and furans, is combined with the phenolic oil to produce bioasphalt binder. In addition to water, the light ends contain a relatively large amount of acetic acid, which can be reacted with limestone or dolomite to produce acetate.[23] Alternatively the light ends can be used in the production of gelled fuel, deicer, or biocement.

The hydrocarbon chemicals portfolio produces aromatics, olefins, and fuel oil. Vispute *et al.*[10] analyzed five different bio-oil upgrading pathways for commodity chemical production from either whole bio-oil or aqueous phase of bio-oil. Here we introduce the hydrocarbon chemicals production pathways through integrated catalytic processing based on their work. Like the bio-based products portfolio, water extraction is used to separate the heavy end fractions (SF1 and SF2) into phenolic oil and solubilized carbohydrates (sugars and anhydrous sugars). Low temperature (21°C), low pressure (1 bar) (LTLP) hydrogenation over 10% Pd/C catalyst has shown to stabilize the phenolic oil to a product with properties similar to traditional No. 2 fuel oil distillate from petroleum.[24] The water soluble fraction along with the middle fraction and light ends (SF3 to SF5) are hydrotreated at 125°C and 100 bar over Ru/C catalyst, followed by a more severe second stage of hydrotreating at 250°C and 100 bar over Pt/C catalyst. Subsequent fluid catalytic cracking at 600°C over HZSM-5 catalyst produces aromatics and olefins. Merchant hydrogen is consumed in hydrogenation and hydrotreating reactors in this portfolio.

Product selection and techno-economic analysis

In a previous study, we separately compared the economic performance of each product portfolio.[15] The same cost analysis method is used and adapted in this work to further explore the economics of the integrated biorefinery by simultaneously producing three product portfolios. IRR, an indicator to measure the break-even point at which the project's net present value equals to zero, is chosen for economic evaluation. A discounted cash flow rate of return (DCFROR) model is employed to calculate IRRs over a 30-year timespan for the integrated pyrolysis biorefinery. Table 1 summarizes the major assumptions and methodology used for cost estimation. All the costs are reported in 2011 US dollars.

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Table 1. Main assumptions and methodology for economic analysis.

Parameter	Assumption	Parameter	Assumption
Plant life	30 years	Total purchased equipment cost (TPEC)	1*TPEC
Equity	40%	Total installed cost (TIC)	3.02*TPEC
Loan interest	7.5%	Indirect cost (IC)	0.89*TPEC
Loan term	10 years	Project contingency	0.78*TPEC
Income tax rate	39%	Fixed capital investment (FCI)	4.69*TPEC
		Working capital (WC)	0.15*TPI
		Land use	0.06*TPEC
		Total project investment (TPI) with land	5.60*TPEC

Table 2. Capital costs of a dedicated biorefinery for specific product portfolios.

	Biofuels ($MM)	Bio-based products ($MM)	Hydrocarbon chemicals ($MM)
Total purchased equipment cost	67.54	35.47	47.06
Pre-treatment and pyrolysis	*27.97*	*27.97*	*27.97*
Fractionation	*2.45*	*2.45*	*2.45*
Upgrading	*12.12*	*1.00*	*12.59*
Hydrogen generation	*20.96*	*0*	*0*
Co-generation	*1.41*	*1.41*	*1.41*
Cooling plant	*0.92*	*0.92*	*0.92*
Balance of plant	*1.72*	*1.72*	*1.72*
Fixed capital investment	316.90	166.41	220.79

The capital costs of individual product portfolios from our previous study are summarized in Table 2, where the differences lie in bio-oil upgrading and hydrogen generation units. In the integrated biorefinery, the bio-oil stream is separated into three fractions towards corresponding product portfolios at the same time instead of flowing into one specific product portfolio. Accordingly, the capital costs of bio-oil upgrading and hydrogen generation are scaled based on the incoming bio-oil fractions towards biofuels, bio-based products, and hydrocarbon chemicals with a power factor of 0.7,[25] while the capital costs of the remaining units are kept the same as each individual one. This approach provides a baseline capital cost for an integrated biorefinery to account for economies of scale. In reality, an integrated biorefinery can incur additional costs due to increased process complexity and unit redundancy which are not considered in this estimation.

The operating costs are calculated by summing the total of material/energy costs and fixed operating costs. Fixed operating costs are broken down into salaries, maintenance, and insurance. A detailed analysis of operating costs of an integrated biorefinery should consider the interaction and integration among material streams to promote the economics of the project. To simplify the analysis, this study calculates the total material/energy costs after giving each material/energy stream a mass flow ratio equivalent to the bio-oil fraction under certain conditions. Table 3 provides the details for material/energy costs calculation which are derived from Aspen Plus™ material and energy balance simulation data and literature values.

The IRR estimates require product selling prices from the biorefinery as an input. We assume the final product prices are the same as their equivalent market commodities because there is limited industrial commercial data for bio-based products. By incorporating all the collected price data into the economic model, we can calculate an IRR over 30-year plant life with a given combination of product yields. Due to price fluctuations of these commodities during the past two decades (from 1994 to 2014), price uncertainty distributions are developed for gasoline, diesel, asphalt binder, cement, dextrose, benzene, and heating oil by fitting available historical monthly prices. Three price levels including low (10th percentile of the distribution), average (mean of the distribution), and high (90th percentile of the distribution) are considered for each product. We assume the products within each product group take the same price level every time for the IRR calculation when considering the price correlation within biofuels, bio-based products, and hydrocarbon chemicals. Thus, a total of 27 price combinations and scenarios are explored.

In order to find the most beneficial operating conditions for the integrated biorefinery, we vary the key parameters, bio-oil fractions flowing into three product portfolios, to compare and select the maximum IRR within one price combination using Mathematica.[26] The same method is then applied to all the scenarios.

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Table 3. Material/energy prices and usage rates for production of biofuels, bio-based products and hydrocarbon chemicals.

Material/energy	Price	Unit	Ref.	Usage			Unit
				Biofuels	Bio-based products	Hydrocarbon chemicals	
Red oak	80	$/MT		2000	2000	2000	MT
Purchased hydrogen	1.5	$/kg	28	0	0	75096.40	kg
Natural gas	0.26	$/kg	15	230523	0	0	lb
Hydrotreating catalyst (Mo-based)	18.25	$/lb	21	1218.43	0	0	lb
Hydrocracking catalyst (Mo-based)	18.25	$/lb	21	852.90	0	0	lb
Hydrotreating catalyst (Ru-based)	2.52	$/lb	14	0	0	545.59	lb
Hydrotreating catalyst (Pt-based)	25.33	$/lb	14	0	0	545.59	lb
Hydrogenation catalyst (Pd-based)	25.33	$/lb	14	0	0	31.60	lb
Hydrocracking catalyst (Zeolite)	0.72	$/lb	14	0	0	186.55	lb
Sulfuric acid	0.08	$/kg	15	0	38893.80	0	kg
Limestone	17	$/MT	15	0	38.47	0	MT
Methanol	400	$/MT	15	0	0	38.51	MT
Biochar	–20	$/MT	15	125.83	125.83	125.83	MT
Hydrogen plant catalyst	4.24	¢/100 kscf H_2	21	34803.50	0	0	kscf
Hydrogen plant process water	195.26	¢/1000 gallon	29	142296	0	0	gallon
Boiler chemicals	1.65	$/lb	30	120	120	120	lb
Cooling tower chemicals	1.18	$/lb	30	17.14	17.14	17.14	lb
Cooling tower makeup	195.26	¢/1000 gallon	29	154008	154621	154621	gallon
Boiling feedwater makeup	195.26	¢/1000 gallon	29	809208	809208	809208	gallon
Sand and ash disposal	0.014	$/lb	28	712138	712138	712138	lb
Waste-water treatment	0.11	$/kg COD	31	794	625	625	kg
Extractor water	195.26	¢/1000 gallon	29	0	180307	180307	gallon
Electricity	6.16	¢/kWh	9	801782	553923	561782	kWh

Uncertainty and sensitivity analysis

Besides evaluating market conditions using 27 product price combinations, we also evaluate and translate the uncertainty of input parameters, product prices, to the uncertainty results of IRR. Stochastic analysis using Monte Carlo simulation is employed. Product prices are randomly generated from their probability distributions for 5000 iterations in IRR calculations. We apply this method to each individual product portfolio first and obtain the IRR probability distributions. Moreover, we take into consideration of the varying bio-oil fractions allocated to three product portfolios in the integrated biorefinery and obtain the IRR probability distribution from random combinations of bio-oil fractions with a total constraint of 100%.

Furthermore, we analyze the effects of other key parameters such as biomass feedstock price, fixed capital cost, and product yields on IRR by conducting a sensitivity analysis. An integrated biorefinery with bio-oil equally allocated for upgrading in three product portfolios (one-third of bio-oil assigned to each product portfolio) and mean product selling prices is considered as an example for the sensitivity analysis.

Results and discussion

Product yields

Product yields for fast pyrolysis and bio-oil upgrading sections are presented in Table 4. Experimental data from fast pyrolysis shows a bio-oil, biochar, and non-condensable gas yield of 65.9wt%, 13.0wt%, and 21.1wt%, respectively. In the integrated biorefinery, NCG and 52% of biochar are burned to provide process heat for the pyrolysis reactor

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Table 4. Biofuels, bio-based products, and hydrocarbon chemicals portfolio yields from fast pyrolysis and bio-oil upgrading.[15]

Input	Output			Unit (MTPD)
Biomass (2000 MTPD)	Biochar			260.80
	Non-condensable gas			421.73
	Bio-oil	SF1		274.25
		SF2		408.29
		SF3		86.82
		SF4		33.01
		SF5		515.10
Bio-oil (1317.47 MTPD)	Biofuels only	Gasoline		266.00
		Diesel		245.00
	Bio-based products only	Asphalt binder		493.68
		Dextrose		197.62
		Calcium acetate		60.84
	Hydrocarbon chemicals only	Aromatics	Benzene	27.77
			Toluene	51.26
			Xylene	20.02
			Ethylbenzene	2.41
			Indene	0.51
			Naphthalene	0.51
			Styrene	1.44
		Olefins	Ethylene	83.30
			Propylene	140.07
			Butylene	36.90
		Fuel oil		287.44

system while excess biochar from pyrolysis counts as credit in calculating net operating costs with a price of $20/MT.

Biofuels, bio-based products, and hydrocarbon chemicals yields from the integrated biorefinery are obtained by scaling product yields in individual product portfolio with bio-oil fractions. If the whole bio-oil is upgraded to produce biofuels, gasoline and diesel yields are 266 and 245 MTPD, which correspond to 96 790 gallons of gasoline and 76 144 gallons of diesel per day. With the entire bio-oil dedicated to bio-based products, we can obtain 493.68 MTPD of asphalt binder, 197.62 MTPD of dextrose, and 60.84 MTPD of calcium acetate. With 100% of bio-oil dedicated to producing hydrocarbon chemicals, we can achieve 101.92 MTPD of aromatics, 260.27 MTPD of olefins, and 287.44 MTPD of fuel oil.

Economics of the integrated biorefinery

The historical price distributions of gasoline, diesel, asphalt binder, cement, dextrose, benzene, and heating oil were fitted to probability distributions as shown in Table 5. The 10th percentile, mean value, and 90th percentile of the product price distributions are taken into IRR calculations. We assume toluene and xylene have the same price distribution as benzene with its mean value shifted based on the literature values where their prices are $1186/MT and $1404/MT.[27] Due to a lack of historical data for a few other hydrocarbon chemicals, we assume ethylene, propylene, and butylene have a fixed price of $1289/MT, 1263/MT, and 1263/MT while ethylbenzene, indene, naphthalene, and styrene have a fixed price of $1356/MT based on the average price of benzene.

We examined a total of 27 scenarios to explore all the combinations of low, average, and high product prices. Under a specific price combination, IRRs are calculated for all possible allocations of bio-oil to biofuels, bio-based products, and hydrocarbon chemicals. An algorithm maximizes the IRRs based on the product portfolio selections using Mathematica's NMaximize function.[26] Table 6 summarizes the bio-oil fractions with the maximum IRR for each scenario.

We found that among the investigated scenarios, most scenarios (Scenarios 4–9, 13–18, and 22–27) tend to maximize IRRs when bio-oil fractions are used for bio-based products production as long as the biorefinery sells the bio-based products at their historical average price or higher. The IRR increases from 42.5% to 67.9% if bio-based products can be sold at their historical high price instead of their historical average price. Biofuels and hydrocarbon chemicals yield lower IRRs than bio-based products unless bio-based products are at historically low market prices.

When switching bio-based product prices to their historical low prices, seven scenarios (Scenarios 1–3, 10–13, and 21) that produce hydrocarbon chemicals generate higher IRRs. Under these conditions, increasing hydrocarbon chemical selling prices from historically low to high prices increases the IRR from 9.1% in Scenario 1 to 27.0% in Scenario 3. An IRR of 9.1% suggests that the capital investment returns would be slightly lower than a common 10% assumption in most economic analysis and possibly result in a monetary loss. With historical low price level of bio-based products, we also found two scenarios (Scenarios 19 and 20) where producing biofuels with their high selling prices yields an IRR of 18.5% even though it is lower than 27.0% from the scenarios of producing hydrocarbon chemicals. This suggests these two scenarios are better choices

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Table 5. Fitted historical price probability distributions and percentiles for commodity products.

Commodity	Probability distribution and parameters	Low (10th percentile)	Average (mean)	High (90th percentile)
Gasoline ($/MT)	Lognormal [μ, α] $\mu = 6.7607$, $\alpha = 0.3207$	572	909	1302
Diesel ($/MT)	Lognormal [μ, α] $\mu = 6.6439$, $\alpha = 0.3598$	484	820	1218
Asphalt binder ($/MT)	Logistic [μ, β] $\mu = 953.651$, $\beta = 244.734$	416	950	1491
Dextrose ($/MT)	Logistic [μ, β] $\mu = 569.491$, $\beta = 76.34$	402	564	737
Cement ($/MT)	Cauchy [a, b] $a = 239.562$, $b = 5.6184$	222	240	257
Benzene ($/MT)	Lognormal [μ, σ] $\mu = 7.2091$, $\mu = 0.07856$	1222	1356	1495
Toluene ($/MT)	Lognormal [μ, σ] $\mu = 7.0753$, $\sigma = 0.07856$	1069	1186	1308
Xylene ($/MT)	Lognormal [μ, σ] $\mu = 7.2440$, $\sigma = 0.07856$	1266	1404	1548
Heating oil ($/MT)	Lognormal [μ, σ] $\mu = 5.8318$, $\sigma = 0.5358$	172	392	678

when the market accepts bio-based products at historically low prices and hydrocarbon chemicals at low or average prices.

Each scenario can be visualized as a ternary plot to find its maximum IRR. A full list of IRR ternary plots for all the scenarios is presented in Figure S1 of the Supporting Information. Figure S2 of the supporting information includes net present value plots based on a 10% IRR for comparison purposes. Figure 2 picks Scenarios 1, 4, 7, and 19 to illustrate the analysis results under certain market situations. The gray spot in the plots represents the maximum IRR of each scenario with a certain combination of bio-oil fractions for biofuels, bio-based products, and hydrocarbon chemicals production. The results indicate the maximum IRR for each scenario always goes to one portfolio instead of a combination of the three. We can see from Scenario 1 that higher hydrocarbon chemicals yields lead to higher IRRs followed by bio-based products yields while producing biofuels is not beneficial as it generates lowest IRR with a negative value. Increasing bio-based products selling prices from a historically low level in Scenario 1 to an average level in Scenario 4, the maximum economic return shifts from producing hydrocarbon chemicals to bio-based products while the worst option remains producing biofuels as indicated by the contour lines. Larger IRRs are observed in Scenario 4 than in Scenario 1. Scenario 7 shows similar trends to Scenario 4 but with higher IRRs and more dense contour lines. This occurs when we raise bio-based products selling prices to a historically high level. Producing biofuels is still the lowest IRR choice under this condition. Scenario 19 is one of two scenarios where producing biofuels presents maximum IRR with its specific price combination. The sparse contour lines indicate that there is limited differentiation in IRR between different product portfolios.

Table 7 summarizes the fixed capital costs, annual operating costs, and maximum IRRs for these four selected scenarios. One hundred percent hydrocarbon chemicals, bio-based products, and biofuels are produced in Scenarios 1, 4 and 7, and 19, respectively. The relatively

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Table 6. Maximum internal rates of return (IRRs) and biofuels, bio-based products, and hydrocarbon chemicals prices and product fractions.

Scenario	Prices			Bio-oil fractions			Maximum IRR
	Biofuels	Bio-based products	Hydrocarbon chemicals	Biofuels	Bio-based products	Hydrocarbon chemicals	
1	Low	Low	Low	0	0	1	9.1%
2	Low	Low	Average	0	0	1	17.5%
3	Low	Low	High	0	0	1	27.0%
4	Low	Average	Low	0	1	0	42.5%
5	Low	Average	Average	0	1	0	42.5%
6	Low	Average	High	0	1	0	41.7%
7	Low	High	Low	0	1	0	67.9%
8	Low	High	Average	0	1	0	67.9%
9	Low	High	High	0	1	0	67.5%
10	Average	Low	Low	0	0	1	9.1%
11	Average	Low	Average	0	0	1	17.4%
12	Average	Low	High	0	0	1	26.8%
13	Average	Average	Low	0	1	0	42.4%
14	Average	Average	Average	0	1	0	42.4%
15	Average	Average	High	0	1	0	40.6%
16	Average	High	Low	0	1	0	67.7%
17	Average	High	Average	0	1	0	67.8%
18	Average	High	High	0	1	0	67.8%
19	High	Low	Low	1	0	0	18.5%
20	High	Low	Average	1	0	0	18.5%
21	High	Low	High	0	0	1	27.0%
22	High	Average	Low	0	1	0	40.3%
23	High	Average	Average	0	1	0	42.5%
24	High	Average	High	0	1	0	42.4%
25	High	High	Low	0	1	0	67.9%
26	High	High	Average	0	1	0	67.8%
27	High	High	High	0	1	0	67.8%

higher capital and operating costs associated with biofuels production contribute to low IRRs for the biorefinery. Lower production costs along with higher market prices make producing bio-based products profitable in most scenarios.

Uncertainty and sensitivity analysis

The probability distributions of IRR for each individual product portfolio are illustrated in Fig. 3. We observe average IRRs of –3.9% for biofuels, 42.7% for bio-based products, and 17.3% for hydrocarbon chemicals respectively with standard deviations of 24.4%, 24.1%, and 6.5% from 5000 runs. It is clear that the biorefinery producing bio-based product alone is most likely to gain positive and larger IRRs even with a higher standard deviation. Hydrocarbon chemicals product portfolio presents a profitable average IRR with a smaller standard deviation, which shows more stable economic advantages and lower risks associated with it. The negative mean value of IRR in biofuels product portfolio along with the high standard deviation indicates that producing biofuels is less attractive and beneficial based on the existing market product prices. The results from uncertainty analysis are highly in line with the previous results using 27 representative price combinations and both of them point out that higher

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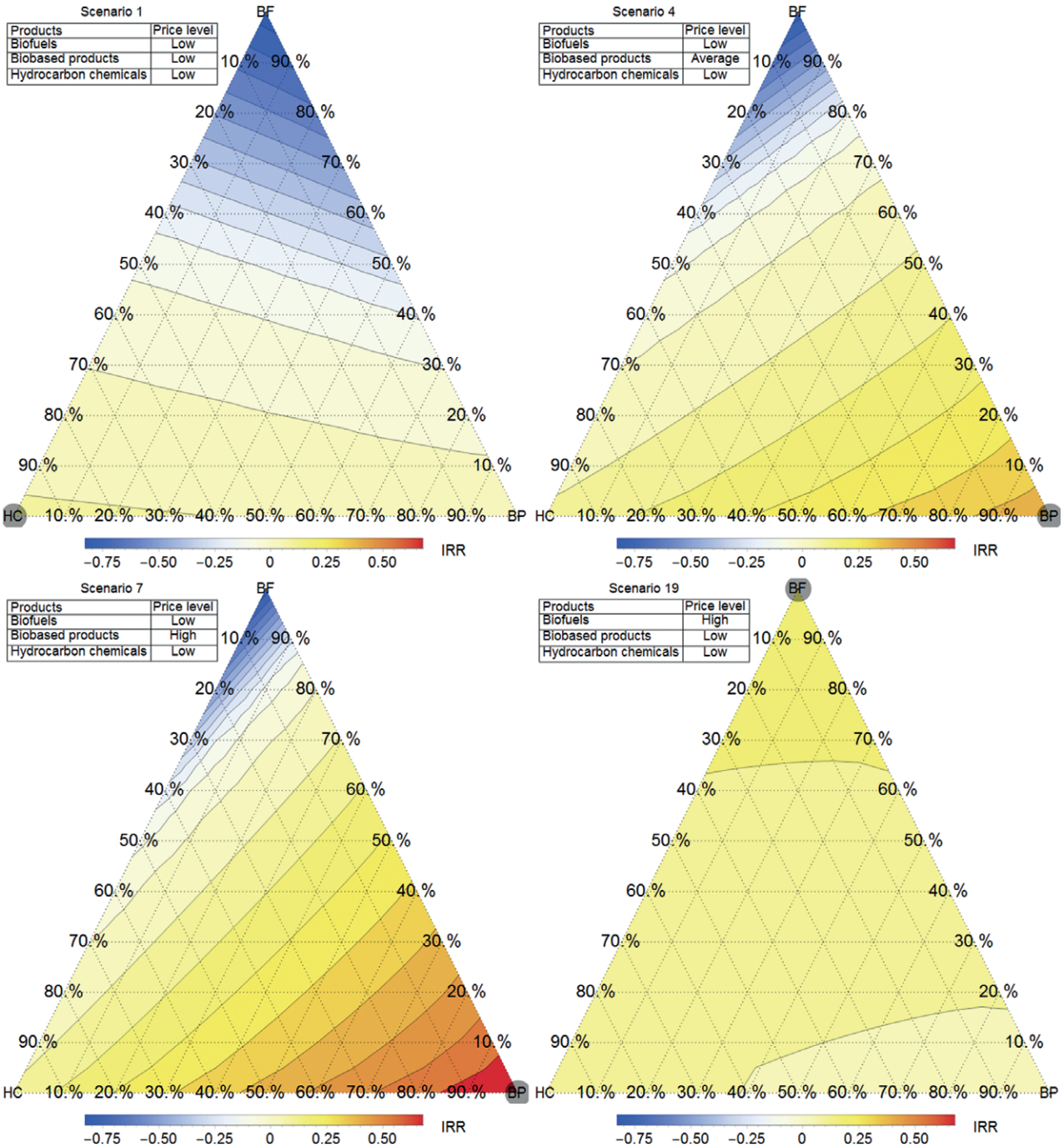

Figure 2. Selected triangle contour plots for internal rate of return (BF: biofuels; BP: bio-based products; HC: hydrocarbon chemicals; IRR: internal rate of return; grey dots: maximum IRR).

Table 7. Fixed capital costs, operating costs, and maximum IRRs for selected scenarios.

	Scenario 1	Scenario 4	Scenario 7	Scenario 19
Fixed capital cost ($MM)	220.8	166.5	166.5	316.9
Annual operating cost ($MM)	129.0	80.6	80.6	122.6
Maximum IRR	9.1%	42.5%	67.9%	18.5%
Product	Hydrocarbon chemicals	Bio-based products	Bio-based products	Biofuels

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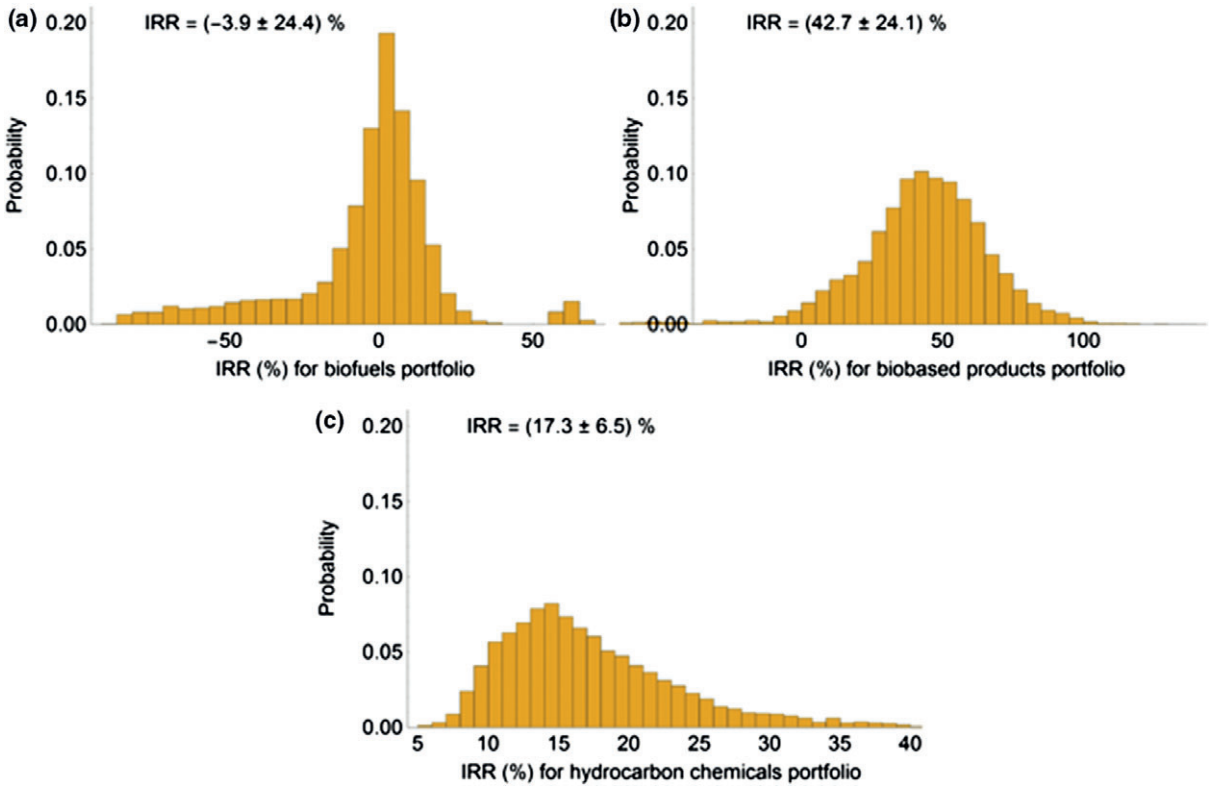

Figure 3. IRR probability distributions for (a) biofuels portfolio, (b) bio-based product portfolio, and (c) hydrocarbon chemicals portfolio.

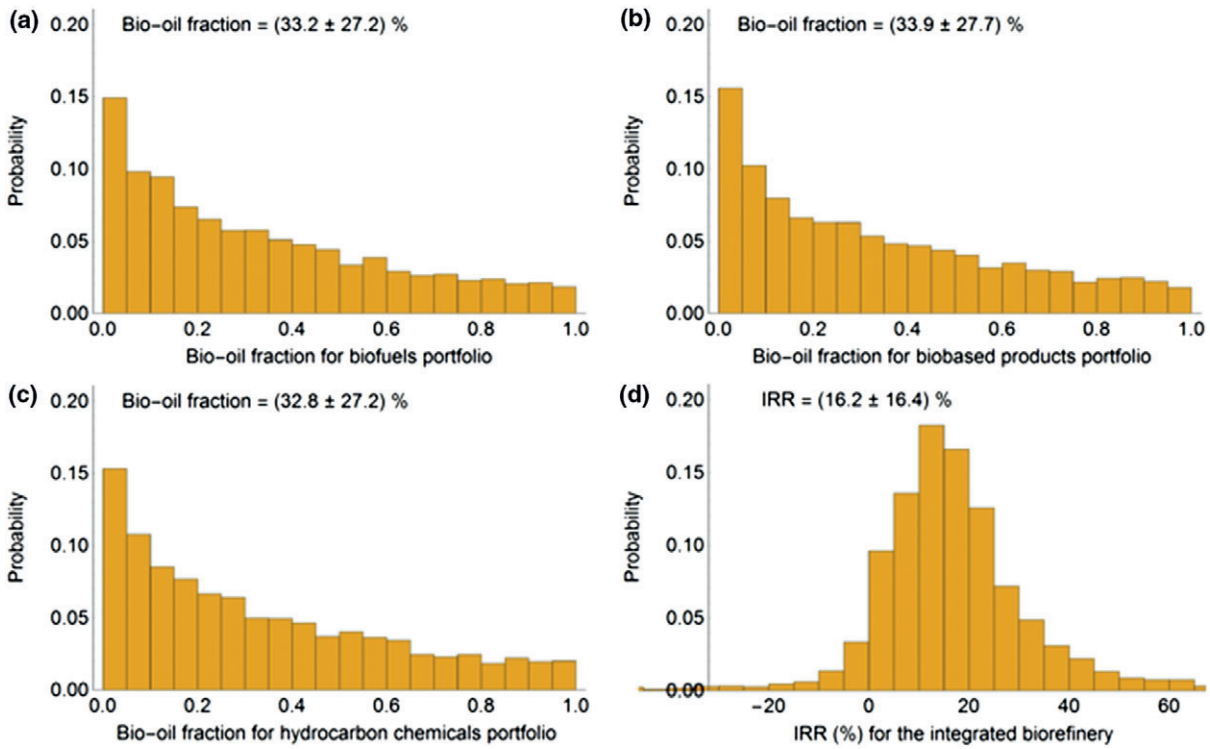

Figure 4. Probability distributions of bio-oil fractions for (a) biofuels portfolio, (b) bio-based product portfolio, (c) hydrocarbon chemicals portfolio, and (d) probability distribution of IRR for the integrated biorefinery.

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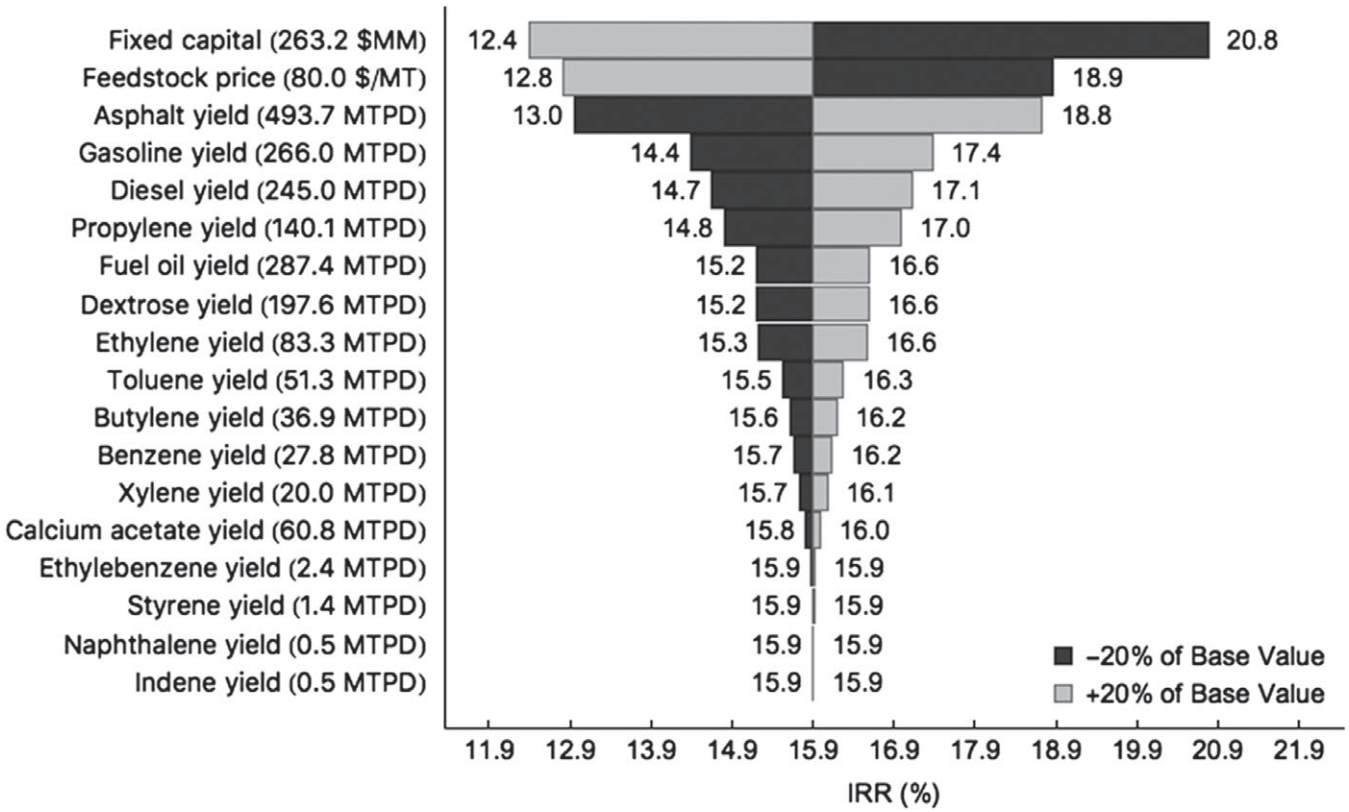

Figure 5. Sensitivity of key parameters on IRR of an integrated biorefinery (bio-oil is equally allocated for upgrading in three product portfolios).

revenues are derived from producing bio-based products under current market conditions.

The probability distribution of IRR for the integrated biorefinery is shown in Fig. 4. This analysis differs from that for individual product portfolio because we also account for the probability distributions of bio-oil fractions allocated for three product portfolios. Since the IRR probability distribution highly depends on distributions of bio-oil fractions, we generate random sets of data for three bio-oil fractions and assume that they are equally likely distributed as shown in Figs 4(a), 4(b), and 4(c). A mean IRR value of 16.2% is observed with a standard deviation of 16.4% for the integrated biorefinery, which indicates a good balance of the disadvantages and advantages of each product portfolio instead of the synergistic effect. These results give investors some insights on diverse product selection and risk mitigation if a better prediction of market conditions is allowed.

In addition to product prices and bio-oil fractions allocated for three product portfolios, we identify the impacts of other important parameters on IRR. The sensitivity results from Fig. 5 illustrate that the fixed capital cost has the largest effect on IRR (an IRR variance of –3.5% to 4.9% with fixed capital cost varying ±20%) followed by the feedstock price. Both parameters show negative impacts that increasing them leads to lower IRRs. On the contrary, all the product yields display positive effects. Among them, asphalt, gasoline, and diesel yields are the significant ones.

Conclusions

Based on the fast pyrolysis and bio-oil fractionation system, three bio-oil upgrading pathways for biofuels, bio-based products, and hydrocarbon chemicals production are considered in this study. We proposed a conceptual pyrolysis biorefinery integrating these product portfolios. The economic performance of the biorefinery is evaluated by calculating and maximizing the internal rates of return for a series of scenarios under different market conditions. The results suggest that producing bio-based products leads to higher IRRs and is the most profitable portfolio for the integrated biorefinery under most historical price levels. The maximum IRR of 67.9% is achieved if bio-based

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products are sold to the market at their historical high prices while an IRR of 42.5% is generated at its historical average selling price. Under low bio-based product market price conditions, biofuels and hydrocarbon chemicals are more profitable with IRRs of 18.5% and 27.0%. These results suggest there are potential economic advantages in optimizing the product allocations of an integrated biorefinery based on market prices.

We also incorporate the uncertainty of input parameters such as product prices and bio-oil fractions allocated for three product portfolios to address the uncertainty distribution of IRR using Monte Carlo simulation. The IRR distributions for biofuels, bio-based products, and hydrocarbon chemicals portfolio give mean values of –3.9%, 42.7%, and 17.3% and standard deviations of 24.4%, 24.1%, and 6.5%, correspondingly. The uncertainty result of the integrated biorefinery displays a mean value of 16.2% with a standard deviation of 16.4%. The results suggest that the biorefinery has higher chances to gain profits by producing bio-based products followed by hydrocarbon chemicals and biofuels. These findings serve as guidance for product selection to reduce the investment risk with known market conditions.

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References

1. US Department of Energy, US Billion-Ton Update: Biomass supply for a bioenergy and bioproducts industry. Report No.: ORNL/TM-2011/224. US DoE, Oak Ridge TN (2011).
2. Ou L, Li B, Dang Q, Jones S, Brown R and Wright MM, Understanding uncertainties in the economic feasibility of transportation fuel production using biomass gasification and mixed alcohol synthesis. *Energy Technol* **4**(3):441–448 (2016).
3. Dang Q, Wright M and Brown RC, Ultra-low carbon emissions from coal-fired power plants through bio-oil co-firing and biochar sequestration. *Environ Sci Technol* **49**(24):14688–14695 (2015).
4. Dang Q, Luo Z, Zhang J, Wang J, Chen W and Yang Y, Experimental study on bio-oil upgrading over Pt/SO42-/ZrO2/SBA-15 catalyst in supercritical ethanol. *Fuel* **103**:683–692 (2013).
5. Bridgwater AV, Review of fast pyrolysis of biomass and product upgrading. *Biomass Bioenergy* **38**:68–94 (2012).
6. Elliott DC, Hart TR, Neuenschwander GG, Rotness LJ and Zacher AH, Catalytic hydroprocessing of biomass fast pyrolysis bio-oil to produce hydrocarbon products. *Environ Prog Sustain Energ* **28**(3):441–449 (2009).
7. Elliott DC, Wang H, Rover M, Whitmer L, Smith R and Brown R, Hydrocarbon liquid production via catalytic hydroprocessing of phenolic oils fractionated from fast pyrolysis of red oak and corn stover. *ACS Sustain Chem Eng* **3**(5):892–902 (2015).
8. Peralta J, Silva HMRD, Williams RC, Rover M and Machado AVA, Development of an innovative bio-binder using asphalt-rubber technology. *Int J Pavement Res Technol* **6**(4):447–456, (2013).
9. Zhang YA, Brown TR, Hu GP and Brown RC, Techno-economic analysis of monosaccharide production via fast pyrolysis of lignocellulose. *Bioresource Technol* **127**:358–365 (2013).
10. Vispute TP, Zhang HY, Sanna A, Xiao R and Huber GW, Renewable chemical commodity feedstocks from integrated catalytic processing of pyrolysis oils. *Science* **330**(6008):1222–1227 (2010).
11. Brown TR and Brown RC, Techno-economics of advanced biofuels pathways. *Rsc Adv* **3**(17):5758–5764 (2013).
12. Wright MM, Daugaard DE, Satrio JA and Brown RC, Techno-economic analysis of biomass fast pyrolysis to transportation fuels. *Fuel* **89**:S2–S10 (2010).
13. Li BY, Ou LW, Dang Q, Meyer P, Jones S, Brown R *et al.*, Techno-economic and uncertainty analysis of in situ and ex situ fast pyrolysis for biofuel production. *Bioresour Technol* **196**:49–56 (2015).
14. Brown TR, Zhang YN, Hu GP and Brown RC, Techno-economic analysis of biobased chemicals production via integrated catalytic processing. *Biofuel Bioprod Bioref* **6**(1):73–87 (2012).
15. Hu W, Dang Q, Rover M, Brown RC and Wright MM, Comparative techno-economic analysis of advanced biofuels, biochemicals, and hydrocarbon chemicals via the fast pyrolysis platform. *Biofuels* **7**(1):87–103 (2015).
16. Zhang YN and Wright MM, Product selection and supply chain optimization for fast pyrolysis and biorefinery system. *Ind Eng Chem Res* **53**(51):19987–19999 (2014).
17. Li YH, Brown T and Hu GP, Optimization model for a thermochemical biofuels supply network design. *J Energ Eng* **140**(4):04014004 (2014).
18. Gebreslassie BH, Slivinsky M, Wang BL and You FQ, Life cycle optimization for sustainable design and operations of hydrocarbon biorefinery via fast pyrolysis, hydrotreating and hydrocracking. *Comput Chem Eng* **50**:71–91 (2013).
19. Pollard AS, Rover MR and Brown RC, Characterization of bio-oil recovered as stage fractions with unique chemical and physical properties. *J Anal Appl Pyrolysis* **93**:129–138 (2012).
20. Rover MR, Johnston PA, Whitmer LE, Smith RG and Brown RC, The effect of pyrolysis temperature on recovery of bio-oil as distinctive stage fractions. *J Anal Appl Pyrolysis* **105**:262–268 (2014).
21. Susanne J, Pimphan M, Snowden-Swan L, Padmaperuma A, Tan E, Dutta A, Jacobson J and Cafferty K, Process design and economics for the conversion of lignocellulosic biomass to hydrocarbon fuels – fast pyrolysis and hydrotreating. PNNL-23053; NREL/TP-5100-61178. US DOE (2013).
22. Rover MR, Johnston PA, Jin T, Smith RG, Brown RC and Jarboe L, Production of clean pyrolytic sugars for fermentation. *ChemSusChem* **7**(6):1662–1668 (2014).
23. Ivanov V and Chu J, Applications of microorganisms to geotechnical engineering for bioclogging and biocementation of soil *in situ*. *Rev Environ Sci Bio/Technol* **7**(2):139–153 (2008).
24. Rover MR, Hall PH, Johnston PA, Smith RG and Brown RC, Stabilization of bio-oils using low temperature, low pressure hydrogenation. *Fuel* **153**:224–230 (2015).

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25. Wright M and Brown RC, Establishing the optimal sizes of different kinds of biorefineries. *Biofuel Bioprod Bioref* **1**(3):191–200 (2007).
26. Wolfram Research Inc., Mathematica. Version 10.3. Wolfram Research, Inc., Champaign, IL (2015).
27. Wang KG, Ou LW, Brown T and Brown RC, Beyond ethanol: a techno-economic analysis of an integrated corn biorefinery for the production of hydrocarbon fuels and chemicals. *Biofuel Bioprod Bioref* **9**(2):190–200 (2015).
28. Wright MM, Satrio JA, Brown RC, Daugaard DE and Hsu David, Techno-economic analysis of biomass fast pyrolysis to transportation fuels. Technical Report No.: NREL/TP-6A20-46586. NREL, Golden, CO (2010).
29. Phillips S, Aden A, Jechura J, Dayton D and Eggeman T, Thermochemical ethanol via indirect gasification and mixed alcohol synthesis of lignocellulosic biomass. Technical Report No.: NREL/TP-510-41168. NREL, Golden, CO (2007).
30. Dutta A and Phillips SD, Thermochemical ethanol via direct gasification and mixed alcohol synthesis of lignocellulosic biomass. Technical Report No.: NREL/TP-510-45913. NREL, Golden CO, (2009).
31. Humbird D, Davis R, Tao L, Kinchin C, Hsu D, Aden A *et al.*, Process Design and economics for biochemical conversion of lignocellulosic biomass to ethanol: dilute-acid pre-treatment and enzymatic hydrolysis of corn stover. Technical Report No.: NREL/TP-510-47764. NREL, Golden CO, (2011).

Dr Qi Dang

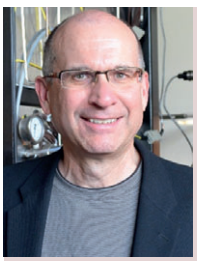
Dr Qi Dang is a Postdoctoral Researcher at the Bioeconomy Institute, Iowa State University. Her research focuses on process modeling, life cycle assessment, and techno-economic analysis of biomass conversion systems.

Wenhao Hu

Wenhao Hu is a Master's student in the Department of Mechanical Engineering and a research assistant at the Bioeconomy Institute, Iowa State University. His research focuses on techno-economic analysis and optimization of biofuels production and bioenergy systems.

Dr Marjorie Rover

Dr Rover is a thermochemical research scientist at the Biorenewable Research Laboratory. Dr Rover supervises the bio-oil analysis laboratory and conducts research in bio-oil characterization and upgrading to chemicals and fuels.



Dr Robert C. Brown

Dr Brown is Anson Marston Distinguished Professor of Engineering and Gary and Donna Hoover Chair in Mechanical Engineering at Iowa State University (ISU). He is the founding director of ISU's Bioeconomy Institute. His research focuses on the thermochemical processing of biomass and fossil fuels into energy, fuels, and chemicals.

Dr Mark M. Wright

Dr Wright is a Black & Veatch Assistant Professor in Mechanical Engineering at Iowa State University. His research interests are in the development and evaluation of biomass conversion technologies using techno-economic analysis, life-cycle assessment, and uncertainty quantification.

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